

Python Notes — Chapter 4: Arrays and Hashing (study material)

Arrays: ordered collections. Access is $O(1)$ by index.

```
nums = [3, 1, 4, 1, 5] # nums[2] is 4
```

Python lists are dynamic arrays: append is amortized $O(1)$.

Hashing: dicts and sets give average $O(1)$ lookup.

```
seen = {} # hash map
```

```
seen['a'] = 1 # insert by hash of key
```

Collisions are resolved internally; never store secrets

in plain text, even in demo code.

Exercises: two-sum with a hash map; group anagrams by sorted key.